FRONT 26 cm.

17 cm.

FINIL 16 mg Tablet

ANGIOTENSIN II RECEPTOR ANTAGONIST

Each tablet contains:
Candesartan cilexetil 16 mg

PRODUCT DESCRIPTION:

Pink, round tablet with engraved CC 16 on one convex side and scored on flat-bevelled side

MECHANISM OF ACTION:

Pharmacology

Angiotensin II is the primary vasoactive hormone of the renin-angiotensin-aldosterone system and plays a role in the pathophysiology of hypertension, heart failure and other cardiovascular disorders. It also has a role in the pathogenesis of end organ hypertrophy and damage. The major physiological effects of angiotensin II, such as vasoconstriction, aldosterone stimulation, regulation of salt and water homeostasis and stimulation of cell growth, are mediated via the type 1 (AT_1) receptor.

Candesartan cilexetil is a prodrug suitable for oral use. It is rapidly converted to the active substance, Candesartan, by ester hydrolysis during absorption from the gastrointestinal tract. Candesartan is an Angiotensin II Receptor Antagonist (AIIRA), selective for AT_1 receptors, with tight binding to and slow dissociation from the receptor. It has no agonist activity.

Candesartan does not inhibit ACE, which converts angiotensin I to angiotensin II and degrades bradykinin. There is no effect on ACE and no potentiation of bradykinin or substance P. Candesartan does not bind to or block other hormone receptors or ion channels known to be important in cardiovascular regulation. The antagonism of the angiotensin II (AT_1) receptors results in dose related increases in plasma renin levels, angiotensin I and angiotensin II levels, and a decrease in plasma aldosterone concentration.

Pharmacokinetics

Absorption and distribution

Following oral administration, Candesartan cilexetil is converted to the active substance Candesartan. The absolute bioavailability of Candesartan is approximately 40% after an oral solution of Candesartan cilexetil. The relative bioavailability of the tablet formulation compared with the same oral solution is approximately 34% with very little variability. The estimated absolute bioavailability of the tablet is therefore 14%. The mean peak serum concentration (C_{max}) is reached 3 to 4 hours following tablet intake. The Candesartan serum concentrations increase linearly with increasing doses in the therapeutic dose range. No gender related differences in the pharmacokinetics of Candesartan have been observed. The area under the serum concentration versus time curve (AUC) of Candesartan is not significantly affected by food.

Candesartan is highly bound to plasma protein (more than 99%). The apparent volume of distribution of Candesartan is 0.1 L/Kg.The bioavailability of Candesartan is not affected by food.

Biotransformation and elimination

Candesartan is mainly eliminated unchanged via urine and bile and only to a minor extent eliminated by hepatic metabolism (CYP2C9). No interaction would be expected to occur with drugs whose metabolism is dependent upon cytochrome P450 isoenzymes CYP1A2, CYP2A6, CYP2C9, CYP2C19, CYP2D6, CYP2E1 or CYP3A4. The terminal half-life of Candesartan is approximately 9 hours. There is no accumulation following multiple doses.

Total plasma clearance of Candesartan is about 0.37 mL/min/Kg, with a renal clearance of about 0.19 mL/min/Kg. The renal elimination of Candesartan is both by glomerular filtration and active tubular secretion. Following an oral dose of ^{14}C-labelled Candesartan cilexetil, approximately 26% of the dose is excreted in the urine as Candesartan and 7% as an inactive metabolite while approximately 56% of the dose is recovered in the feces as Candesartan and 10% as the inactive metabolite.

Pediatric population

In children aged 1 to < 6 years, 10 children weighing 10 to < 25 Kg received a single dose of 0.2 mg/Kg, oral suspension.

There was no correlation between C_{max} and AUC with age or weight. No clearance data has been collected; therefore, the possibility of a correlation between clearance and weight/age in this population is unknown.

In children aged 6 to < 17 years, there is no correlation between C_{max} and AUC with age. However, weight seems to significantly correlate with C_{max} (p = 0.012) and AUC (p = 0.011). The possibility of a correlation between clearance and weight/age in this population is unknown.

Children > 6 years of age had exposure similar to adults given the same dose.

The pharmacokinetics of Candesartan cilexetil have not been investigated in pediatric patients < 1 year of age.

INDICATION**:

- Treatment of hypertension
- Treatment of patients with heart failure and impaired left ventricle systolic function (left ventricular ejection fraction ≤ 40%) as add-on therapy to ACE inhibitors or when ACE inhibitors are not tolerated.

DOSAGE AND ADMINISTRATION**:

Oral

Dosage in Hypertension

The recommended initial dose and usual maintenance dose is 8 mg once daily. The dose may be increased to 16 mg once daily. If blood pressure is not sufficiently controlled after 4 weeks of treatment with 16 mg once daily, the dose may be further increased to a maximum of 32 mg once daily (see Pharmacodynamic properties). If blood pressure control is not achieved with this dose, alternative strategies should be considered.

Therapy should be adjusted according to blood pressure response. Most of the antihypertensive effect is attained within 4 weeks of initiation of treatment.

Use in the elderly

No initial dosage adjustment is necessary in elderly patients.

Use in patients with intravascular volume depletion

An initial dose of 4 mg may be considered in patients at risk for hypotension, such as patients with possible volume depletion (see Warning and precaution).

Use in impaired renal function

No initial dosage adjustment is necessary in patients with mild to moderate renal impairment (i.e creatinine clearance 30-80 mL/min/1.73 m^2 BSA). In patients with severe renal impairment (i.e. creatinine clearance < 30 mL/min/1.73 m^2 BSA), the clinical experience is limited and a lower initial dose of 4 mg should be considered.

Use in impaired hepatic function

Patient with hepatic impairment: Dose titration is recommended in patients with mild to moderate chronic liver disease, and a lower initial dose of 4 mg should be considered. Finil 16 should not be used in patients with severe hepatic impairment and/or cholestasis (see Contraindication).

Concomitant therapy

Addition of a thiazide-type diuretic such as hydrochlorothiazide has been shown to have an additive antihypertensive effect with Finil 16.

Use in black patients

The antihypertensive effect of Candesartan is less in black than non-black patients. Consequently, uptitration of Finil 16 and concomitant therapy may be more frequently needed for blood pressure control in black than non-black patients (see also Pharmacodynamics properties).

Dosage in Heart Failure

The usual recommended initial dose of Candesartan is 4 mg once daily. Upitration to the target dose of 32 mg once daily or the highest tolerated dose is by doubling the dose at intervals of at least 2 weeks (see section Warning and precaution).

Special patient populations

No initial dose adjustment is necessary for elderly patients or in patients with intravascular volume depletion, renal impairment or mild to moderate hepatic impairment.

Concomitant therapy

Finil 16 can be administered with other heart failure treatment, including ACE inhibitors, beta-blockers, diuretics and digitalis or a combination of these medicinal products (see also section Warning and Precaution and Pharmacodynamic properties).

Use in children and adolescents

The safety and efficacy of Finil 16 have not been established in children and adolescents (under 18 years).

Finil 16 should be taken once daily with or without food.

WARNING AND PRECAUTION**:

Dual blockade of the renin-angiotensin-aldosterone system (RAAS)

There is evidence that the concomitant use of ACE inhibitors, angiotensin II receptor blockers or Aliskiren increases the risk of hypotension, hyperkalemia and decreased renal function (including acute renal failure). Dual blockade of RAAS through the combined use of ACE-inhibitors, angiotensin II receptor blockers or Aliskiren is therefore not recommended (see section Drug interaction and Pharmacodynamics).

If dual blockade therapy is considered absolutely necessary, this should only occur under specialist supervision and subject to frequent close monitoring of renal function, electrolytes and blood pressure.

ACE-inhibitors and angiotensin II receptor blockers should not be used concomitantly in patients with diabetic nephropathy.

Renal impairment

As with other agents inhibiting the renin-angiotensin-aldosterone system, changes in renal function may be anticipated in susceptible patients treated with Finil 16.

When Finil 16 is used in hypertensive patients with renal impairment, periodic monitoring of serum potassium and creatinine levels is recommended. There is limited experience in patients with very severe or end-stage renal impairment ($Cl_{creatinine}$ < 15 mL/min). In these patients Finil 16 should be carefully titrated with thorough monitoring of blood pressure.Evaluation of patients with heart failure should include periodic assessments of renal function, especially in elderly patients 75 years or older, and patients with impaired renal function. During dose titration of Finil 16, monitoring of serum creatinine and potassium is recommended.

Use in pediatric patients including patients with renal impairment

Finil 16 has not been studied in children with a glomerular filtration rate less than 30 mL/min/1.73 m^2 (see section Dosage and administration).

Concomitant therapy with an ACE inhibitor in heart failure

The risk of adverse reactions, especially hypotension, hyperkalemia and decreased renal function (including acute renal failure), may increase when Finil 16 is used in combination with an ACE inhibitor (see section Side effects). Triple combination of an ACE-inhibitor, a mineralocorticoid receptor antagonist and Candesartan cilexetil is also not recommended. Use of these combinations should be under specialist supervision and subject to frequent close monitoring of renal function, electrolytes and blood pressure.

ACE-inhibitors and angiotensin II receptor blockers should not be used concomitantly in patients with diabetic nephropathy.

Hemodialysis

During dialysis the blood pressure may be particularly sensitive to AT1-receptor blockade as a result of reduced plasma volume and activation of the renin-angiotensin-aldosterone system. Therefore, Finil 16 should be carefully titrated with thorough monitoring of blood pressure in patients on hemodialysis.

Renal artery stenosis

Medicinal products that affect the renin-angiotensin-aldosterone system, including angiotensin II receptor antagonists (AIIRAs), may increase blood urea and serum creatinine in patients with bilateral renal artery stenosis or stenosis of the artery to a solitary kidney.

Kidney transplantation

There is no experience regarding the administration of Finil 16 in patients with a recent kidney transplantation.

Hypotension

Hypotension may occur during treatment with Finil 16 in heart failure patients. It may also occur in hypertensive patients with intravascular volume depletion such as those receiving high dose diuretics. Caution should be observed when initiating therapy and correction of hypovolemia should be attempted. For children with possible intravascular volume depletion (e.g. patients treated with diuretics, particularly those with impaired renal function), Candesartan treatment should be initiated under close medical supervision and a lower starting dose should be considered (see section Dosage and administration).

Anesthesia and surgery

Hypotension may occur during anesthesia and surgery in patients treated with angiotensin II antagonists due to blockade of the renin-angiotensin system. Very rarely, hypotension may be severe such that it may warrant the use of intravenous fluids and/or vasopressors.

Aortic and mitral valve stenosis (obstructive hypertrophic cardiomyopathy)

As with other vasodilators, special caution is indicated in patients suffering from hemodynamically relevant aortic or mitral valve stenosis, or obstructive hypertrophic cardiomyopathy.

Primary hyperaldosteronism

Patients with primary hyperaldosteronism will not generally respond to antihypertensive medicinal products acting through inhibition of the renin-angiotensin-aldosterone system. Therefore, the use of Finil 16 is not recommended in this population.

Hyperkalemia

Concomitant use of Finil 16 with potassium-sparing diuretics, potassium supplements, salt substitutes containing potassium, or other medicinal products that may increase potassium levels (e.g. Heparin) may lead to increases in serum potassium in hypertensive patients. Monitoring of potassium should be undertaken as appropriate.

In heart failure patients treated with Finil 16, hyperkalemia may occur. Periodic monitoring of serum potassium is recommended. The combination of an ACE inhibitor, a potassium-sparing diuretic (e.g. Spironolactone) and Finil 16 is not recommended and should be considered only after careful evaluation of the potential benefits and risks.

General

In patients whose vascular tone and renal function depend predominantly on the activity of the renin-angiotensin-aldosterone system (e.g. patients with severe congestive heart failure or underlying renal disease, including renal artery stenosis), treatment with other medicinal products that affect this system has been associated with acute hypotension, azotemia, oliguria or, rarely, acute renal failure. The possibility of similar effects cannot be excluded with AIIRAs. As with any antihypertensive agent,

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	FINIL 16	IFMY 0040	W 26 x L 17 cm.	Wood Free Paper (กระดาษปอนด์)	60 g (0.08 mm.)
	Designed by: (PDM, ASPD, PDC-M, PDC-A, PDO-M, PDO-A)	Checked by: (PDM/ASPD)		Approved by: (MD (Only Pattern and Colors))	
Approved by:	PLC: (Date)	LRA: (Date)	IRA: (Date)	QCC-PM: (Date)	CUSTOMER/SALES DEPT.: (Date)

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excessive blood pressure decrease in patients with ischemic cardiopathy or ischemic cerebrovascular disease could result in a myocardial infarction or stroke.
The antihypertensive effect of Candesartan may be enhanced by other medicinal products with blood pressure lowering properties, whether prescribed as an antihypertensive or prescribed for other indications.
Finil contains lactose. Patients with rare hereditary problems of galactose intolerance total lactase deficiency or glucose-galactose malabsorption should not take this medicine.

Pregnancy

AIIRAs should not be initiated during pregnancy. Unless continued AIIRA therapy is considered essential, patients planning pregnancy should be changed to alternative antihypertensive treatments which have an established safety profile for use in pregnancy. When pregnancy is diagnosed, treatment with AIIRAs should be stopped immediately, and, if appropriate, alternative therapy should be started (see sections Contraindication and Pregnancy and lactation).
In post-menarche patients the possibility of pregnancy should be evaluated on a regular basis. Appropriate information should be given and/or action taken to prevent the risk of exposure during pregnancy (see sections Contraindication and Pregnancy and lactation).

Warning:
This preparation contains Sodium metabisulphite that may cause serious allergic type reactions in certain susceptible patients. Do not use if known to be hypersensitive to bisulphites.

PREGNANCY AND LACTATION:

Pregnancy

The use of AIIRAs is not recommended during the first trimester of pregnancy. The use of AIIRAs is contraindicated during the second and third trimesters of pregnancy.

Epidemiological evidence regarding the risk of teratogenicity following exposure to ACE inhibitors during the first trimester of pregnancy has not been conclusive; however a small increase in risk cannot be excluded. While there is no controlled epidemiological data on the risk with AIIRAs, similar risks may exist for this class of drugs.
Unless continued AIIRA therapy is considered essential, patients planning pregnancy should be changed to alternative antihypertensive treatments which have an established safety profile for use in pregnancy. When pregnancy is diagnosed, treatment with AIIRAs should be stopped immediately and, if appropriate, alternative therapy should be started.
Exposure to AIIRA therapy during the second and third trimesters is known to induce human fetotoxicity (decreased renal function, oligohydramnios, skull ossification retardation) and neonatal toxicity (renal failure, hypotension, hyperkalemia).Should exposure to AIIRAs have occurred from the second trimester of pregnancy, ultrasound check of renal function and skull is recommended.Infants whose mothers have taken AIIRAs should be closely observed for hypotension.

Breastfeeding

Because no information is available regarding the use of Candesartan cilexetil during breastfeeding, Candesartan cilexetil is not recommended and alternative treatments with better established safety profiles during breastfeeding are preferable, especially while nursing a newborn or preterm infant.

SIDE EFFECT**:

Treatment of Hypertension

The overall incidence of adverse events showed no association with dose or age. The most commonly reported adverse reactions were dizziness/vertigo, headache and respiratory infection.

System Organ Class	Frequency	Side Effect
Infections and infestations	Common	Respiratory infection
Blood and lymphatic system disorders	Very rare	Leukopenia, neutropenia and agranulocytosis
Metabolism and nutrition disorders	Very rare	Hyperkalemia, hyponatremia
Nervous system disorders	Common	Dizziness/vertigo, headache
Respiratory, thoracic and mediastinal disorders	Very rare	Cough
Gastrointestinal disorders	Very rare	Nausea
	Not known	Diarrhea
Hepato-biliary disorders	Very rare	Increased liver enzymes, abnormal hepatic function or hepatitis
Skin and subcutaneous tissue disorders	Very rare	Angioedema, rash, urticaria, pruritus
Musculoskeletal and connective tissue disorders	Very rare	Back pain, arthralgia, myalgia
Renal and urinary disorders	Very rare	Renal impairment, including renal failure in susceptible patients (see section Warning and precaution)

Laboratory findings

In general, there were no clinically important influences of Finil 16 on routine laboratory variables. As for other inhibitors of the renin-angiotensin-aldosterone system, small decreases in hemoglobin have been seen. No routine monitoring of laboratory variables is usually necessary for patients receiving Finil 16. However, in patients with renal impairment, periodic monitoring of serum potassium and creatinine levels is recommended.

Pediatric population

Whilst the nature and severity of the adverse events are similar to those in adults (see Table above), the frequency of all adverse events are higher in children and adolescent, particularly in:

- Headache, dizziness and upper respiratory tract infection
- Cough
- Rash
- Hyperkalemia, hyponatremia and abnormal liver function
- Sinus arrhythmia, nasopharyngitis, pyrexia and oropharyngeal pain.However, these are temporary and widespread childhood illnesses.

The overall safety profile for Candesartan cilexetil in paediatric patients does not differ significantly from the safety profile in adults.

Treatment of Heart Failure

The adverse experience profile of Finil 16 in adult heart failure patients was consistent with the pharmacology of the drug and the health status of the patients. The most commonly reported adverse reactions were hyperkalemia, hypotension and renal impairment. These events were more common in patients over 70 years of age, diabetics, or subjects who received other medicinal products which affect the renin-angiotensin-aldosterone system, in particular an ACE inhibitor and/or Spironolactone.

System Organ Class	Frequency	Side Effect
Blood and lymphatic system disorders	Very rare	Leukopenia, neutropenia and agranulocytosis
Metabolism and nutrition disorders	Common	Hyperkalemia
	Very rare	Hyponatremia
Nervous system disorders	Very rare	Dizziness, headache
Vascular disorders	Common	Hypotension
Respiratory, thoracic and mediastinal disorders	Very rare	Cough
Gastrointestinal disorders	Very rare	Nausea
	Not known	Diarrhea
Hepato-biliary disorders	Very rare	Increased liver enzymes, abnormal hepatic function or hepatitis
Skin and subcutaneous tissue disorders	Very rare	Angioedema, rash, urticaria, pruritus
Musculoskeletal and connective tissue disorders	Very rare	Back pain, arthralgia, myalgia
Renal and urinary disorders	Common	Renal impairment, including renal failure in susceptible patients (see section Warning and precaution)

Laboratory findings

Hyperkalemia and renal impairment are common in patients treated with Finil 16 for the indication of heart failure. Periodic monitoring of serum creatinine and potassium is recommended (see section Warning and precaution).

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES:

No studies on the effects of Candesartan on the ability to drive and use machines have been performed. However, it should be taken into account that occasionally dizziness or weariness may occur during treatment with Candesartan cilexetil.

CONTRAINDICATION:

- Hypersensitivity to Candesartan cilexetil or to any of the excipients
- Second and third trimester of pregnancy
- Severe hepatic and/or cholestasis
- Children aged below 1 year

The concomitant use of Candesartan cilexetil with Aliskiren-containing products is contraindicated in patients with diabetes mellitus and renal impairment (GFR < 60 mL/min/1.73 m^2).

DRUG INTERACTION:

Compounds which have been investigated in clinical pharmacokinetic studies include Hydrochlorothiazide, Warfarin, Digoxin, oral contraceptives (i.e. Ethinyl estradiol/ Levonorgestrel), Glibenclamide, Nifedipine and Enalapril. No clinically significant pharmacokinetic interactions with these medicinal products have been identified.Concomitant use of potassium-sparing diuretics, potassium supplements, salt substitutes containing potassium, or other medicinal products (e.g. Heparin) may increase potassium levels. Monitoring of potassium should be undertaken as appropriate.
Reversible increases in serum Lithium concentrations and toxicity have been reported during concomitant administration of Lithium with ACE inhibitors. A similar effect may occur with AIIRAs. Use of Candesartan with Lithium is not recommended. If the combination proves necessary, careful monitoring of serum Lithium levels is recommended.
When AIIRAs are administered simultaneously with non-steroidal anti-inflammatory drugs (NSAIDs) (i.e. selective COX-2 inhibitors, Acetylsalicylic acid (> 3 g/day) and non-selective NSAIDs), attenuation of the antihypertensive effect may occur.
As with ACE inhibitors, concomitant use of AIIRAs and NSAIDs may lead to an increased risk of worsening of renal function, including possible acute renal failure, and an increase in serum potassium, especially in patients with poor pre-existing renal function. The combination should be administered with caution, especially in the elderly. Patients should be adequately hydrated and consideration should be given to monitoring renal function after initiation of concomitant therapy, and periodically thereafter.

OVERDOSE AND TREATMENT:

Symptoms

Based on pharmacological considerations, the main manifestation of an overdose is likely to be symptomatic hypotension and dizziness. In individual case reports of overdose (of up to 672 mg Candesartan cilexetil) in an adult, patient recovery was uneventful.

Management

If symptomatic hypotension should occur, symptomatic treatment should be instituted and vital signs monitored.
The patient should be placed supine with the legs elevated. If this is not sufficient, plasma volume should be increased by infusion of, for example, isotonic saline solution. Sympathomimetic medicinal products may be administered if the above-mentioned measures are not sufficient.
Candesartan is not removed by hemodialysis.

SHELF LIFE:

Three (3) years

STORAGE:

Store at temperature of not more than 30°C.

DOSAGE FORM AND PACKAGING AVAILABLE:

16 mg Tablet, Blister 10x10's

DATE OF REVISION:

December 19, 2019

Manufactured by:
UNISON LABORATORIES CO., LTD.
39 Moo 4, Klong Udomcholjorn, Muang Chachoengsao,
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Product Registration Holder/ Importer:
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11400 Ayer Itam, Penang Malaysia

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IFMY 0040

PM Specification

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Checked by: (PDM/ASPD)
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